Farbod Siahkali

Curriculum Vitae | October 2026

Electrical Engineering, Purdue University | West Lafayette, Indiana

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Education

Ph.D. in Electrical Engineering (Automatic Control Area), Purdue University, Aug. 2024 - May 2029 (Expected), West Lafayette, IN, USA. Advisor: Prof. Vijay Gupta.

M.S. in Electrical Engineering, Purdue University, Expected December 2026 (GPA: 3.83/4.00), West Lafayette, IN, USA.

B.S. in Electrical Engineering (Control Engineering Branch), University of Tehran, Sep. 2019 - July 2023 (18.36/20), Tehran, Iran.

Research Interests

Conformal prediction and uncertainty quantification; multi-sensor fusion and reliable perception; trajectory optimization and multi-agent planning; domain adaptation; safe reinforcement learning and control barrier functions.

Research Experience

Graduate Research Assistant, Prof. Vijay Gupta's Lab, Purdue University, Aug. 2024 - Present

- Develop reliable perception and planning methods using conformal prediction, multi-sensor fusion, trajectory optimization, and safe reinforcement learning.
- Developed conformal late fusion for dependent sensor outputs. Experiments on synthetic data, CIFAR-10, CIFAR-100, and MELD reduced prediction-set size by **10-25% at fixed coverage**.
- Developed slack-inflated and source-tuned pseudo-calibration algorithms for domain adaptation, with Wasserstein-based coverage bounds under label-conditional covariate shift.
- Developed single- and multi-agent deferred-decision trajectory optimization using mixed-integer conic optimization, conformal prediction, sequential candidate rejection, and receding-horizon planning.
- Developed barrier-continuation policy gradient (BCPG) for safe reinforcement learning with online parameter estimation and control barrier functions, evaluated in adaptive cruise control and planar navigation.

Research Assistant, Intelligent Networks Lab, University of Tehran, Oct. 2023 - July 2024

- Implemented deep deterministic policy gradient (DDPG) policies for networked epidemic control and budget-constrained opinion shaping.

Research Assistant, Human & Robot Interaction Laboratory (TaarLab), University of Tehran, May 2021 - Oct. 2022

- Developed deep convolutional neural networks for person re-identification, attribute recognition, and attribute retrieval.
- Built PyTorch pipelines with YOLOv5 and OSNet for detection, re-identification, and multi-object tracking.
- Created CA-Duke labels for 32,000+ images and contributed to the SIVD dataset with 36,000+ images and 29 classes.

Peer-Reviewed Publications

Farbod Siahkali, Mengxue Hou, Vijay Gupta. **Active Sensing and Deferred-Decision Trajectory Optimization for Robust Target Identification.** IEEE Control Systems Letters (L-CSS), vol. 10, pp. 811-816, 2026. Published. CDC 2026 presentation accepted.

https://doi.org/10.1109/LCSYS.2026.3703893

Farbod Siahkali, Vijay Gupta. **Conformal Prediction-Based Multi-Sensor Classification.** Neurocomputing, vol. 678, article 133182, 2026. Published.

https://doi.org/10.1016/j.neucom.2026.133182

Saba Samadi, **Farbod Siahkali**, Philip E. Paré. **Deep Reinforcement Learning for Epidemic Control of a Networked SIS Model.** IEEE Conference on Control Technology and Applications (CCTA), pp. 145-150, 2025. Published.

https://ieeexplore.ieee.org/document/11151322

Farbod Siahkali, Seyed Amirmahdi Alavi, Mehdi Tale Masouleh. **SIVD: Dataset of Iranian Vehicles for Real-Time Multi-Camera Video Tracking and Recognition.** 2022 8th Iranian Conference on Signal Processing and Intelligent Systems (ICSPIS), pp. 1-7. Published.

https://ieeexplore.ieee.org/document/10043932

Manuscripts Under Review

Farbod Siahkali (first author). **Multi-Agent Deferred-Decision Trajectory Optimization with Anytime-Valid Target Retention for Moving Candidates.** Under review at IEEE Robotics and Automation Letters (RA-L).

Farbod Siahkali, Ashwin Verma, Vijay Gupta. **Coverage Guarantees for Pseudo-Calibrated Conformal Prediction under Distribution Shift.** Under review at IEEE Signal Processing Letters. Public preprint: arXiv:2602.14913 (2026).

https://arxiv.org/abs/2602.14913

Preprints

Farbod Siahkali, Saba Samadi, Hamed Kebriaei. **Towards Opinion Shaping: A Deep Reinforcement Learning Approach in Bot-User Interactions.** arXiv:2409.11426, 2024. Preprint.

https://arxiv.org/abs/2409.11426

Hossein Bodaghi, **Farbod Siahkali**, Shayan Samiee, Seraj Ghasemi, Mehdi Tale Masouleh, Ahmad Kalhor. **Image-Based and Partially Categorical Annotating Approach for Pedestrian Attribute Recognition.** SSRN, 2023. Preprint.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4373086

Work in Preparation

Farbod Siahkali (first author). **Policy Gradient with Online Parameter Estimation and Evolving CBF Action Sets.** In preparation for submission to IEEE Control Systems Letters (L-CSS).

Teaching Experience

Teaching Assistant, University of Tehran, 2020-2023

Neural Networks & Deep Learning (Master's course), Spring & Fall 2023 - Dr. Kalhor.

Operational Research, Fall 2023 - Dr. Ramezaney Moghadam.

Instrumentation, Fall 2022 & Spring 2023 - Dr. Nayeri & Dr. Nasiri.

Linear Control Systems, Fall 2022 - Dr. Bahrami.

Engineering Mathematics, Fall 2021 - Dr. Tale Masouleh; Spring 2021 - Dr. Taheri.

Introduction to Computing Systems and Programming, Fall 2020 & Fall 2021 - Dr. Moradi.

Mentoring

Research Mentor, Purdue ASPIRE Research Experience for Undergraduates (REU), 2026.

Academic Service

Reviewer

Neurocomputing.

IEEE Control Systems Letters (L-CSS).

IEEE Conference on Decision and Control (CDC).

International Conference on Intelligent Computing and Next Generation Networks (ICNGN 2023).

International Conference on Electrical, Computer, Communications, & Mechatronics Engineering (ICECCME 2024).

International Conference on Electrical and Computer Engineering Researches (ICECER 2024).

International Conference on Electrical, Computer, and Energy Technologies (ICECET 2024).

Honors & Awards

Best Undergraduate Project Award, ECE Faculty, University of Tehran, July 2023.

Technical Skills

Languages & Tools: Python, MATLAB, C, Git, Linux, LaTeX.

ML Libraries: PyTorch, TensorFlow, NumPy, Pandas, SciPy, scikit-learn.

Methods: Conformal prediction, uncertainty quantification, sensor fusion, Bayesian inference, domain adaptation, mixed-integer/conic optimization, trajectory planning, multi-agent systems, safe reinforcement learning, control barrier functions, computer vision.

Selected Graduate Coursework

Neural Networks & Deep Learning; Deep Generative Models; Reinforcement Learning; Robotics & Mechatronics; Nonlinear Control Systems; Game Theory; Lumped System Theory.